

Origins of juvenile Woodcock (*Scolopax rusticola*) harvested in Spain inferred from stable hydrogen isotope ($\delta^2\text{H}$) analyses of feathers

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Abstract Woodcock (*Scolopax rusticola*) are heavily hunted in south-central Europe during the winter and have been the focus of intense management efforts and concern. One of the significant challenges with this and other migratory gamebirds is that they are typically produced in locations other than those where they are primarily harvested. The delineation of origins of harvested Woodcock is important for the identification of appropriate management strategies. Using analysis of stable hydrogen isotope ratios ($\delta^2\text{H}$) in feathers of juvenile Woodcock harvested in Spain (2007–2009, $n = 802$ individuals), we assigned Woodcock origins using a derived Woodcock feather $\delta^2\text{H}$ isoscape and four potential isotopic regions covering the breeding range. We also used a continuous probability surface approach that used the forest cover available throughout the range as a constraint on origins. The majority (90 %) of Woodcock captured in Spain were assigned to origins in central-eastern Europe (488 individuals, 60 %), and the Baltic region (237 individuals, 30 %). These results were similar to those previously obtained for Woodcock taken in France (although the relative importance of both eastern Europe and Baltic regions of origin was the opposite to those harvested in Spain). This means that a significant portion of the Spanish and

French Woodcock bags can be considered as the same management stock, and thus that management of hunter takes in both countries should be coordinated. Additionally, our results provide further evidence suggesting the need for conservation of forests in Eastern Europe, the southern Baltic, the Ukraine, Belarus, and western Russia.

Keywords Deuterium · Feathers · Harvest · Migratory connectivity · *Scolopax rusticola* · Stable isotopes · Woodcock

Zusammenfassung

Herkunft von in Spanien erjagten juvenilen Waldschnepfen (*Scolopax rusticola*), abgeleitet aus der Analyse stabiler Wasserstoffisotope ($\delta^2\text{H}$) in Federn

Waldschnepfen (*Scolopax rusticola*) werden in Südeuropa im Winter intensiv bejagt, was Anlass zur Besorgnis gibt, und stehen im Mittelpunkt intensiver Managementbemühungen. Eine der wesentlichen Herausforderungen im Umgang mit dieser Art sowie anderen jagdbaren Zugvögeln besteht darin, dass sie typischerweise an anderen Orten gejagt als „produziert“ werden. Eine Beschreibung der Herkunft erjagter Waldschnepfen ist wichtig, um geeignete Managementstrategien zu ermitteln. Mittels einer Analyse der Verhältnisse stabiler Wasserstoffisotope ($\delta^2\text{H}$) in Federn juveniler in Spanien erjagter Waldschnepfen (2007–2009, $n = 802$ Individuen) haben wir die Herkunft der Vögel unter Verwendung einer abgeleiteten Waldschnepfen-Feder- $\delta^2\text{H}$ -Isotopenlandschaft sowie vier potenzieller Isotopenregionen, die das Brutgebiet abdecken, zugeordnet. Wir haben außerdem einen kontinuierlichen Wahrscheinlichkeitsoberflächen-Ansatz benutzt, der die Herkunft der Vögel auf der Basis der Waldbedeckung im

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Brutgebiet einschränkte. Dem Großteil (90 %) der in Spanien gefangenen Waldschnepfen wurde eine Herkunft aus Zentral-Osteuropa (488 Individuen, 60 %) und dem Baltikum (237 Individuen, 30 %) zugeordnet. Diese Ergebnisse ähnelten denen, die zuvor für in Frankreich erjagte Waldschnepfen gewonnen worden waren, wobei dort mehr Individuen aus dem Baltikum und weniger Individuen aus Osteuropa stammten. Dies bedeutet, dass ein erheblicher Anteil der spanischen und französischen Waldschnepfen-Jagdbeute wohl aus demselben Management-Bestand stammt und das Management der Jagdquote in den beiden Ländern koordiniert werden sollte. Außerdem liefern unsere Ergebnisse weitere Hinweise für die Notwendigkeit des Schutzes von Wäldern in Osteuropa, dem südlichen Baltikum, der Ukraine, Weißrussland und Russland.

Introduction

In response to considerable limitations of more conventional mark–recapture approaches, the last decade has seen the development of a number of new tools to track the origins and movements of migratory animals (Hobson and Norris 2008; Bowen et al. 2005). One such tool involves the use of intrinsic markers in animal tissues, removing the need for initial marking. In particular, the measurement of naturally occurring stable isotope ratios of hydrogen ($\delta^2\text{H}$) in animal tissues to infer origins relative to known or expected long-term patterns of $\delta^2\text{H}$ in rainfall at continental scales, has revolutionized the way in which animals can be tracked (Hobson and Wassenaar 2008). This approach has been particularly successful for the study of migratory birds, since feathers can lock in isotopic signals following growth and molt phenology is well known for most species. Recently, it has been recognized that this approach is particularly well suited for the conservation and management of migratory game birds, since it is vitally important to link regions of production with regions of harvest. To date, this approach has been used in North America to investigate the origins of hunter-killed Sandhill Crane (*Grus canadensis*; Hobson et al. 2006), Lesser Scaup (*Aythya affinis*; Hobson et al. 2009a), and American Black Duck (*Anas rubripes*; Ashley et al. 2010). In Europe, the approach has successfully delineated the origins of hunter-killed woodpigeon (*Columba palumbus*, Hobson et al. 2009b) in France as well as the origins of Mallard (*Anas platyrhynchos*) infected with avian influenza virus moving through southern Sweden (Gunnarson et al. 2012). Most recently, we have applied the isotope approach to linking origins of juvenile Woodcock (*Scolopax rusticola*) hunted throughout France in winter (Hobson et al. 2006, and in

review). That investigation highlighted the importance of forests in central and eastern Europe and western European Russia as important production zones for Woodcock wintering in France.

Woodcock is an important game species in Spain during winter, but there is relatively little information about its ecology, demography or connections with other populations (Lucio and Sáenz de Buruaga 2000) which would be important for developing adaptive strategies for managing this species and enhancing sustainable use. The linkage between populations wintering in France and Spain has been already highlighted because France may be responsible for taking an important proportion of birds migrating through France to winter in Spain, and Woodcock wintering in France may move to Spain during periods of cold weather (Gossmann and Ferrand 2000; Guzmán et al. 2010; Peron et al. 2011). It would also be important to determine whether the origins of birds wintering in France and Spain are similar, as this would also mean that both bags have to be managed as the same stock.

Previous studies of the origin of Woodcock wintering in Spain have been based on relatively few recoveries of ringed birds (Guzmán et al. 2010). Those analyses suggest that birds wintering in Spain mostly breed in the circum-Baltic region including western Russia. Both Sweden and western Russia have been assumed to be particularly important. These recoveries also suggest that there may be a differential distribution of birds within Spain using different migration strategies, with long-distance migrants wintering further south, consistent with a leapfrog movement as proposed by Ferrand and Gossmann (2009a).

Here, our primary objective was to delineate origins of Woodcock from the Spanish hunt through the analysis of feathers grown on natal grounds. We also evaluated whether timing of arrival or harvest location within Spain was related to origin, and make recommendations for the conservation and management of Woodcock wintering in southwestern Europe.

Methods

Flight feathers were recovered from 802 juvenile Woodcocks taken in Spain in the autumn migration period ($n = 286$; October–30 November), during the wintering period ($n = 492$; 1 December–20 February) and the spring migration period ($n = 24$, 21 February–March) in the 2007–2008 and 2008–2009 autumn–winters. Feathers were collected from birds shot by hunters (Club de Cazadores de Becada) during the winter hunting season and caught by ringers of the Woodcock Network (CCB/Txepetxa/Forest Wardens of Guipúzcoa/ABN) in autumn–winter. Feathers were collected in 21 provinces of continental Spain, mainly

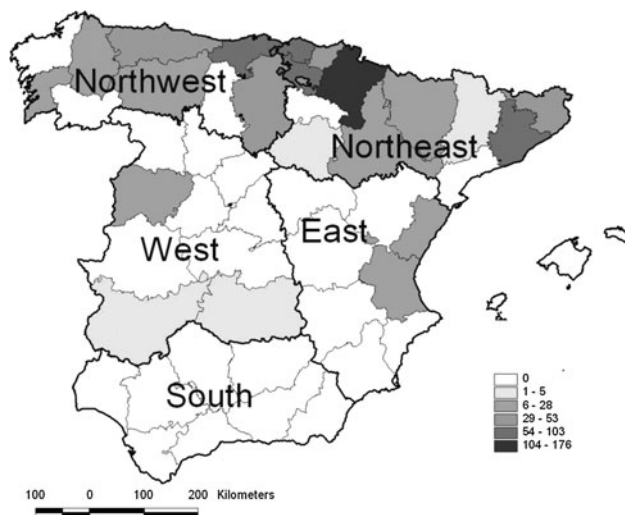


Fig. 1 Provinces in Spain where Woodcock (*Scolopax rusticola*) samples were collected in autumn migrations of 2007–2008 and 2008–2009, and regions used for analyses

in the northern areas (Fig. 1a). Samples were assigned to one of five main regions in Spain (Fig. 1b): northeastern (NE), northwestern (NW), eastern (E), western (W), or southern (S). No samples were collected for the southern region.

The first secondary feather was selected for isotope analysis. Juvenile Woodcock do not molt flight feathers before migration (Ferrand and Gossmann 2009b), thus sampled feathers were grown in the natal grounds.

Stable isotope analysis

Feather samples were stored in paper envelopes and subsequently analyzed at the Environment Canada stable isotope facility in Saskatoon, Canada. Following several months of equilibration with ambient air in Saskatoon, the samples were cleaned using a 2:1 chloroform:methanol rinse and air-dried in a fume hood. We corrected for exchangeable hydrogen by conducting hydrogen isotope analyses using the comparative equilibration method described by Wassenaar and Hobson (2003, 2006) using three calibrated keratin hydrogen isotope reference materials. Isotopic measurements were performed on H_2 gas derived from high-temperature (1,350 °C) flash pyrolysis (Eurovector elemental analyser, Milan, Italy) of $350 \pm 10 \mu g$ feather subsampled from the distal section of the feather vanes and keratin standards using continuous-flow isotope-ratio mass spectrometry (Isoprime, Manchester, UK). Measurements of the three keratin laboratory reference materials (CFS, CHS, BWB) corrected for linear instrumental drift were both accurate and precise with typical within-run mean $\delta^2H \pm SD$ values of $-147.4 \pm 0.79 \text{ ‰}$ ($n = 5$), $-187 \pm 0.56 \text{ ‰}$ ($n = 5$), and $-108 \pm$

0.33 ‰ ($n = 5$), respectively. A control keratin reference yielded a 6-month running SD of $\pm 3.3 \text{ ‰}$ ($n = 76$). All results are reported for non-exchangeable H and are expressed in the typical delta notation, in units of per mil (‰), and normalized on the Vienna Standard Mean Ocean Water Standard Light Antarctic Precipitation (VSMOW-SLAP) standard scale.

Statistical analysis

We used linear models to explore the causes of variation in feather δ^2H (δ^2H_f). We considered models including region of capture (NE, NW, E, W), year and season (migration, winter) as factors. Our candidate set of models included an intercept-only null model, and models with combinations of the main effects. We selected amongst competing models using Akaike's information criterion adjusted for small sample size (AIC_c) and AIC model weights to rank our candidate models from most to least parsimonious (Burnham and Anderson 2002). We considered models within two AIC_c units of the top model as competitive.

For samples collected during autumn migration, we also considered models including year as a factor, date as a linear covariate (with 1 = 1 October), and their interaction to test for differences in arrival dates according to origin.

Assignment of Woodcock to geographic origins

Based upon analysis of δ^2H_f , we assigned Woodcock to their approximate geographic origins using likelihood-based assignment procedures. We first converted a GIS-based isoscape of mean annual δ^2H in precipitation (δ^2H_p , Bowen et al. 2005), based upon regression δ^2H_f from known-origin Woodcock in Europe (Powell 2012) against δ^2H_p using linear modeling. The linear model included δ^2H_p as a linear covariate, and age class (juvenile versus adult) as a factor. We used the resulting parameter estimates for juvenile birds to recalibrate the Bowen et al. (2005) isoscape using the equation $\delta^2H_f = -24.22 + 0.80 \delta^2H_p$. The resulting Woodcock feather isoscape was then further delimited using a digital layer of the breeding range (digitized from Snow and Perrins 1998) using a Spatial Analyst™ “extract values by mask” operation. The resulting isoscapes were then exported in ascii raster format for all subsequent analyses.

We conducted likelihood-based assignment to origin by using normal probability density functions to estimate the likelihood that any given cell within the δ^2H_f isoscape represented a potential origin for an individual by comparing the observed δ^2H_f value with that predicted by the isoscape (Hobson et al. 2009b; Van Wilgenburg and Hobson 2011). The normal probability density functions were parameterized by treating the isoscape predicted δ^2H_f

value at each cell as the mean, and the standard deviation (11.7 ‰) in the residuals from the calibration equation of Powell (2012).

Similar to Hobson et al. (2006, and in review), we further restricted our assignments to origin by incorporating estimates of the proportion of forest cover within the cells of the isoscape as prior probabilities in the likelihood assessments using Baye's Theorem (Royle and Rubenstein 2004; Hobson et al. 2009b; Van Wilgenburg and Hobson 2011). We derived estimates of forest cover from the European forest cover (GIS) database (http://www.efi.int/portal/virtual_library/information_services/mapping_services/forest_map_of_europe/) which combines data derived from the IRS, SPOT, and AVHRR satellites aggregated to a 1-km resolution (Kempeneers et al. 2011; Päivinen et al. 2001; Schuck et al. 2002). These data were subsequently reprojected to a WGS-84 projection and resampled to a resolution of 0.33° to match our $\delta^2\text{H}_f$ isoscape. We derived four bins approximately corresponding to natal origins in Spain and France, central and eastern Europe, the Baltic region (including southwestern Russia), and northern Fennoscandia (including northwestern Russia) (Fig. 2). Our isotopic bins were selected to maximize between-bin differences in mean feather $\delta^2\text{H}$ values predicted from calibrating the $\delta^2\text{H}_p$ isoscape of Bowen et al. (2005) to allow statistical separation of potential origins given the expected level of variance between individuals (above). We assigned

individual birds to one of these four bins by summing the estimated probability of origins across cells falling within these regions using a GIS “zonal” statistics query, and assigned individuals into the bin for which the highest cumulative likelihood was obtained.

We also conducted spatially explicit assignments to origin by selecting the raster cells consistent with the upper 67 % of estimated “probabilities of origin” for each individual (consistent with 2:1 odds of being correct) and coded these as 1 and coded all others cells as 0 (Hobson et al. 2009b; Van Wilgenburg and Hobson 2011; Gunnarsson et al. 2012). This resulted in one binary map (1 = likely origin, 0 = unlikely origin) per individual, and we summed the individual assignments over all individuals by addition of the individual binary surfaces to depict the origins of the entire sample (Hobson et al. 2009b; Van Wilgenburg and Hobson 2011; Gunnarsson et al. 2012). All assignments and associated GIS operations were completed using functions written within the R statistical computing environment (R Development Core Team 2012) using the ‘raster’ package (Hijmans and van Etten 2012).

Results

Four of the models we considered to explain variation in $\delta^2\text{H}_f$ were within 2 AIC_c units of the top model, and one of them included the null (intercept only) model (Table 1). Our top model suggested that $\delta^2\text{H}_f$ varied among samples collected within different regions of Spain; however, this model only explained ~1 % of the variation in $\delta^2\text{H}_f$. Samples collected in the western region (mean = −74 ‰, SD = 14 ‰) were depleted in ^2H by approximately 5 ‰ relative to samples collected in other regions of Spain (mean = −69 ‰, SD = 12 ‰; Fig. 3a). Our second best model also suggested that, in addition to regional

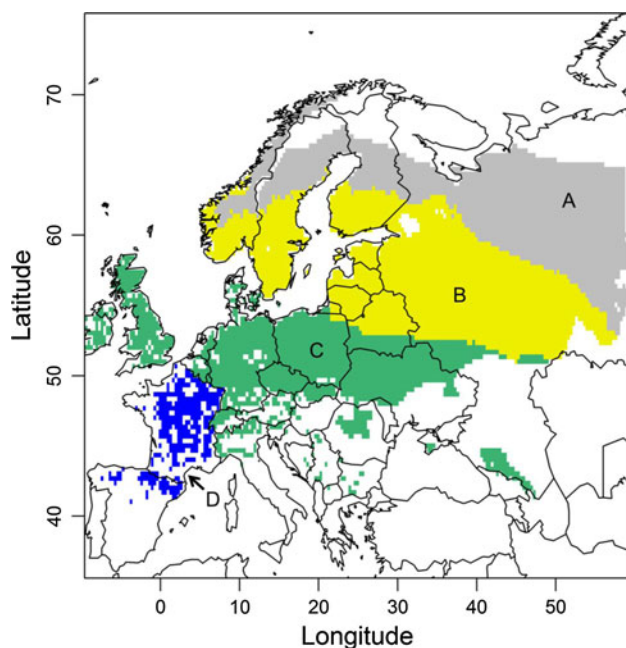


Fig. 2 Geographic range to which Woodcock assignments to origin were conducted using isotopic regions for a categorical (binning) approach (see “Methods”). Isotopic regions are referred to in the text as northern Fennoscandia/northwestern Russia (a), the Baltic/western Russia (b), central Europe (c), and Spain/France (d)

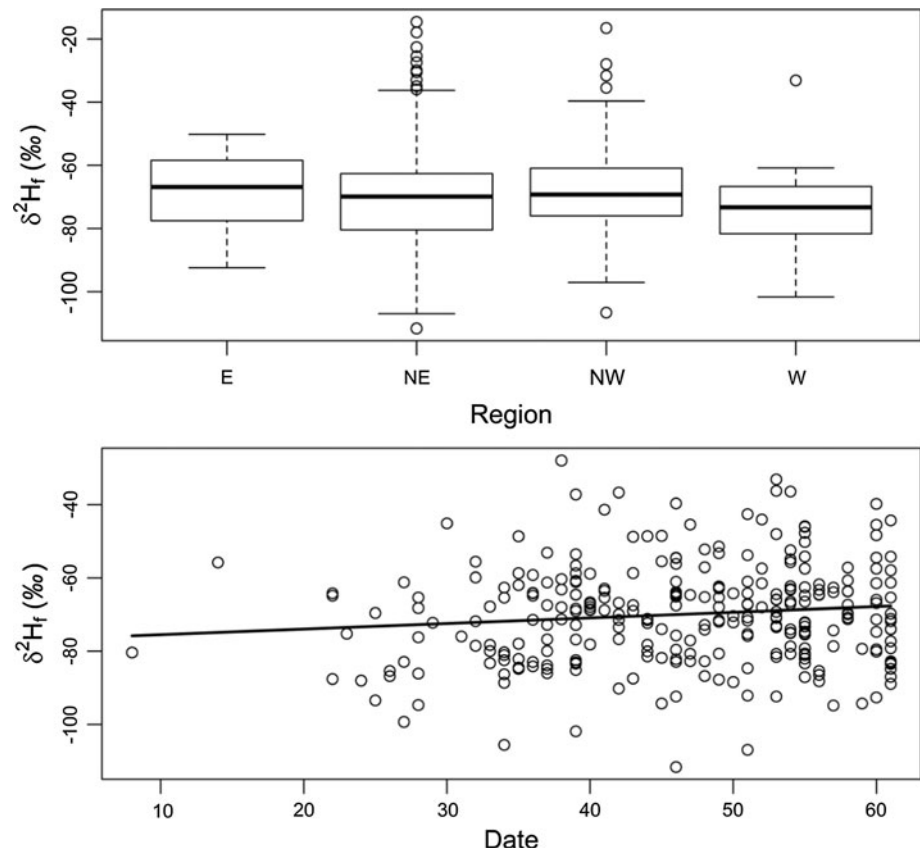
Table 1 Model confidence set (models within two AIC_c units of the top model) for linear models exploring variation in $\delta^2\text{H}_f$ for juvenile Woodcock (*Scolopax rusticola*) sampled in Spain in the non-breeding season of 2007–2008 and 2008–2009

Model	K	AIC_c	ΔAIC_c	ω_i	Cum. ω_i
Zone	5	6,480.0	0.00	0.31	0.31
Zone + year	6	6,481.0	0.89	0.20	0.51
Null (intercept only)	2	6,481.4	0.91	0.20	0.71
Year	3	6,481.9	1.42	0.15	0.87
Season + zone	6	6,482.2	1.73	0.13	1.00

Model selection was assessed by ranking model parsimony based upon Akaike's information criterion adjusted for small sample size (AIC_c) and AIC model weights (Burnham and Anderson 2002)

K denotes the number of estimated parameters, ΔAIC_c is AIC relative the most parsimonious model, ω_i is the AIC weight, and Cum. ω_i is the cumulative weight

Fig. 3 Stable hydrogen isotope values of Woodcock collected in different regions in Spain in winter (*upper* regions as in Fig. 1) and in different dates during autumn migration (*lower* where 1 = 1 October), 2007–2009. Regression line in lower panel predicts progressive enrichment in ^2H through time [$y = -77.04 \text{ ‰}$ (3.68 SE) + 0.15 (0.08 SE) (asterisk) Date]



differences, there were differences in $\delta^2\text{H}_f$ among sampling years, but the confidence interval for year overlapped zero and differences between years were trivial (2007–2008 mean = -70 ‰ , SD = 13.5 ‰ ; and other -69 ‰ , SD = 13.1 ‰). The remaining alternative models (Table 1) were outcompeted by the null (intercept only) model, and thus were not considered further.

When examining variation in $\delta^2\text{H}_f$ from birds caught during autumn migration, two models were within 2 AIC_c units of the top model including the null model (Table 2). The top model included date as an explanatory variable whereas the second most-parsimonious model contained both date and year as parameters. Confidence intervals for the year parameter in the second best model overlapped zero, thus we restricted all inference to the model only including date as a linear covariate. Our top model explained $\sim 1 \%$ of the variance in $\delta^2\text{H}_f$ and suggested that migrants showed enrichment in ^2H through time ($\beta = 0.15$, 0.08 SE). Thus, birds taken at the end of migration (Day 60) were enriched in ^2H by approximately 9 ‰ relative to birds sampled at the beginning of migration (see Fig. 3b), suggesting that birds from northern areas arrived earlier.

The majority (90 %) of Woodcock captured in Spain were assigned to origins in central Europe and the Baltic

Table 2 Model confidence set (models within two AIC_c units of the top model) for linear models exploring variation in $\delta^2\text{H}_f$ for juvenile Woodcock sampled in Spain in autumn migrations of 2007–2008 and 2008–2009

Model	K	AIC_c	ΔAIC_c	ω_i	Cum ω_i
Date	3	2,308.6	0.00	0.52	0.52
Date + year	4	2,309.9	1.26	0.28	0.80
Null (intercept only)	2	2,310.5	1.86	0.20	1.00

Model selection was assessed by ranking model parsimony based upon Akaike's information criterion adjusted for small sample size (AIC_c) and AIC model weights (Burnham and Anderson 2002)

K denotes the number of estimated parameters, ΔAIC_c is AIC relative to the most parsimonious model, ω_i is the AIC weight, and Cum. ω_i is the cumulative weight

regions (Table 3; Fig. 4). The largest proportion of samples were associated with central Europe (488 individuals, 61 %), with another 30 % of samples (237 individuals) coming from the Baltic region. Only 67 birds (8 %) were potential residents from Spain or birds coming from France, and fewer still (10 birds) were likely to have originated from northern Fennoscandia and northwestern Russia (Table 3). Combining birds assigned to northern Fennoscandia and the Baltic as long-distance migrants, there was a greater proportion of birds that were

Table 3 Assigned origins of 802 juvenile Woodcock sampled in Spain in 2007–2008 and 2008–2009, showing the % of Woodcock sampled in each Spanish region assigned to each isotopic region (sample size in parentheses)

Sampling region	Assigned region			
	A	B	C	D
Migration (autumn and spring)				
East	0 (0)	31 (4)	62 (8)	8 (1)
Northeast	2 (6)	32 (76)	57 (138)	9 (21)
Northwest	0 (0)	29 (15)	59 (30)	12 (6)
West	0 (0)	20 (1)	60 (3)	20 (1)
Total	2 (6)	31 (96)	58 (179)	9 (29)
Winter				
East	0 (0)	21 (3)	71 (10)	7 (1)
Northeast	1 (2)	32 (97)	58 (174)	9 (28)
Northwest	1 (1)	22 (35)	72 (116)	6 (9)
West	6 (1)	38 (6)	56 (9)	0 (0)
Total	1 (4)	29 (141)	63 (309)	8 (38)
All seasons				
East	0 (0)	26 (7)	67 (18)	7 (2)
Northeast	1 (8)	32 (173)	58 (312)	9 (49)
Northwest	0 (0)	24 (50)	69 (146)	7 (15)
West	5 (1)	33 (7)	57 (12)	5 (1)
Total	1 (10)	30 (237)	61 (488)	8 (67)

Isotopic regions (see Fig. 1) are referred to in the text as northern Fennoscandia/north-central Russia (A), the Baltic/western Russia (B), central Europe (C), France and Spain (D)

long-distance migrants in autumn ($\sim 45\%$) and spring ($\sim 49\%$) than during winter ($\sim 37\%$), but this difference was not statistically significant ($\chi^2 = 1.91$, $df = 2$, $p > 0.1$). Similarly, the proportion of long-distance migrants in the western region of Spain ($\sim 38\%$), was greater than in the other regions ($\sim 31\%$), and this difference was even more marked in the winter sample (44 vs. 29 %); however, these differences were not statistically significant (Fisher's Exact test, $p > 0.1$).

Discussion

Our isotopic analysis of feathers of juvenile Woodcock grown on the natal grounds indicates that most individuals derived from forests of central to eastern Europe, followed by forests of the southern Baltic and southwestern Russia. Few birds were assigned as coming from northern Fennoscandia and northwestern Russia.

These results are partly consistent with the analysis of ring recoveries by Guzmán et al. (2010), which suggested that most Spanish wintering birds derived from the circum-Baltic regions. However, the predominance of assignments to isotopic region C (central and eastern Europe) was

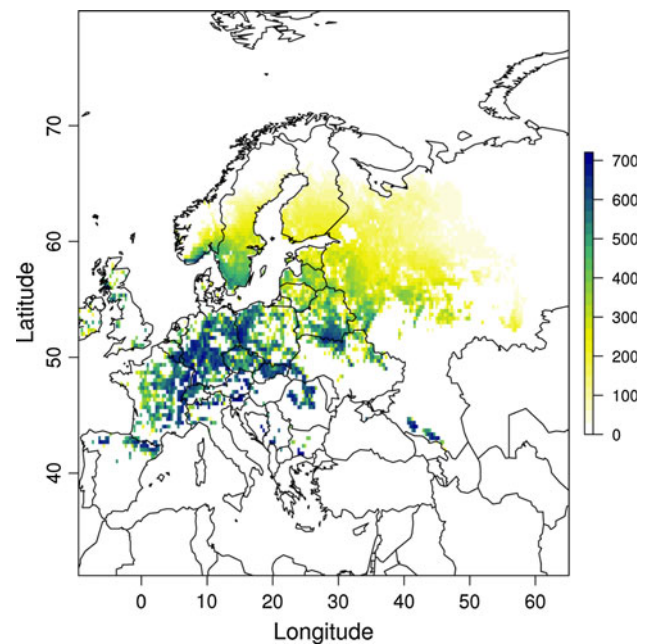


Fig. 4 Potential origin of 802 juvenile Woodcock sampled in Spain in autumn migrations of 2007–2008 and 2008–2009. Legend scale indicates the number of individuals that were assessed as being isotopically consistent with the map pixel representing a likely origin for the specified number of individuals using our likelihood of origin approach to assignment (see “Methods”)

unexpected, and in contrast with results based on ringing recoveries (Guzmán et al. 2010), and indicates a stronger importance of more southern areas as the origin of birds wintering in Spain than previously expected. These differences may be partly due to the fact that countries where Woodcock ringing campaigns are non-existent or scarce may be underestimated as possible origins in ringing recoveries analyses. This could be the case for many of the countries of the isotopic region of central-eastern Europe, such as Poland, Ukraine, and Belarus, where there are no ringing campaigns during the breeding season, or Germany, Austria, the Netherlands, or the Czech Republic where the ringing effort is not reported (Guzmán et al. 2010). Additionally, because we lacked sufficient knowledge of the spatially explicit relative breeding density of Woodcock throughout Europe (BirdLife International 2004), we were unable to use this as a prior to better constrain our assignments (Royle and Rubenstein 2004; Hobson et al. 2009a). Among countries included within the “central-eastern Europe” isotopic region, Woodcock breeding populations in Austria, Czech Republic, the Netherlands, Belgium, Switzerland, or Hungary are relatively small (BirdLife International 2004), and ringing recoveries analyses strongly indicate that birds from the United Kingdom or Denmark seldom migrate to Spain (Guzmán et al. 2010). Thus, these countries likely contribute very little to the Spanish wintering stock even

though they are predicted to be isotopically similar to wintering birds sampled in Spain. Germany is also included in the isotopic region C, and hosts an important breeding population of Woodcock. However, there are no recoveries of German birds in Spain, although ringing effort has not been quantified (Guzmán et al. 2010). Thus, it is likely that most birds identified as coming from isotopic region C come primarily from Poland, southern Belarus, or northern Ukraine.

Only about 8 % of our samples originated in France or Spain. Another isotope study has shown that the proportion of resident birds among the wintering stock in France was lower than 6 % (Hobson et al. 2006, and in review), and thus many of the French birds likely migrate further south or alternatively are numerically overwhelmed by migrants from further north. Additionally, <1 % of Spanish samples were assigned to an isotopic region including Spain and southern France. Overall, these results indicate that the Spanish hunted population needs to be recognized as being almost entirely migratory.

Our analysis of temporal variation in isotopic values during migration provides only weak support for the hypothesis that birds from further north arrived in Spain earlier in the autumn, possibly as a consequence of shorter breeding seasons or colder temperatures there. According to the leapfrog migration strategy proposed by Ferrand and Gossmann (2009a), we had anticipated that birds wintering in southwestern areas of Spain would show a greater proportion of Woodcock originating from more northern regions, and that birds wintering in Spain would have a higher proportion of northern birds as compared to France. Birds taken in the southwestern extreme of our sampling area contained a higher proportion of birds derived from further north. Woodcock hunting in southern Spain is much less common than further north and so sample sizes were small. It is possible that the latitudinal gradient we examined (e.g., France to Spain) was not large enough to detect leapfrog migration if it exists using $\delta^2\text{H}_f$. A more rigorous isotopic test of the leapfrog hypothesis will be to examine birds wintering in more central portions of their wintering range and compare those to the French and Spanish populations that constitute the most southwestern portion of the range.

Our results indicate an overall similar catchment area of birds wintering in Spain as compared to Woodcock harvested in France, with both samples being centered around -71 ‰ (Hobson et al. 2006, and in review; this study), and both including a small proportion of resident birds within the wintering population. This means that a significant portion of the Spanish Woodcock bag can be considered as the same management stock as that of France, and thus a coordinated hunting management plan between both countries would be beneficial. Finally, our results provide

additional evidence suggesting the need for emphasis on conservation of forests in eastern Europe, the southern Baltic area, the Ukraine, Belarus, and western Russia. Again, it is important to emphasize here that these depictions show the isotopic solution space of probable origins for birds and not the actual origins per se. While we cannot confirm these forests as being important source areas for the French and Spanish harvests, the larger forest tracts in eastern Europe, the southern Baltic, and western Russia fall into this high probability region, and those forests are likely to be contributing more recruits to the hunted populations in Spain and France. Additionally, our results also provide additional impetus to reconsider the aggregate contributions of more fragmented forests across central Europe.

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